

Density fluctuation statistics and turbulence spreading at the edge of L-mode plasmas

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Abstract

Long-wavelength density fluctuations ($k\rho_i < 1$) are studied using beam emission spectroscopy (BES) at the edge of DIII-D L-mode plasmas ($\rho = 0.88\text{--}1.1$) in scenarios with electron cyclotron heating (ECH) power ramp (P_{ECH} up to 1.5 MW), neutral beam injection (NBI) power ramp (P_{NBI} up to 2.5 MW), and injected torque scan ($-1 < T_{\text{inj}} < 0.6$ Nm). We find that broadband turbulent density fluctuations ($f \sim 20\text{--}120$ kHz) have a non-Gaussian distribution. The skewness of $\delta n/n$ changes sign from negative at $\rho < 0.95\text{--}0.97$ to positive at $\rho > 0.97$, indicating the prevalence of density ‘voids’ at inner radii and density ‘blobs’ at outer radii and outside of the separatrix. The turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ is calculated to characterize turbulence spreading at the plasma edge. During ECH/NBI power ramps and at counter- I_p injected torque, $\langle \tilde{v}_r \tilde{n}^2 \rangle$ is directed inward inside the separatrix, which is evidence of inward spreading of turbulence intensity from the edge gradient region caused by the inner propagation of density ‘voids’. Significantly weaker $\langle \tilde{v}_r \tilde{n}^2 \rangle$ is observed with co- I_p torque. A correlation between co- I_p torque, turbulence intensity $\delta n/n$ at $\rho = 0.97$, and increased scrape-off layer (SOL) heat flux decay length λ_q is found in the torque scan scenario, showing that edge turbulence plays a material role in determining the SOL conditions and heat flux width.

Keywords: tokamak, edge density fluctuations, turbulence spreading

(Some figures may appear in colour only in the online journal)

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1. Introduction

Recent experiments on tokamaks and stellarators, as well as global gyrokinetic simulations, show that local turbulence characteristics are not solely determined by the local plasma parameters such as temperature and density profile gradients or local transport coefficients [1–4]. Turbulence can spread from the regions where it is excited to the regions where it is locally stable, or where its level is lower, for instance from the plasma edge to the scrape-off layer (SOL), leading to an increase in the turbulence intensity in the SOL. Turbulence spreading from the plasma edge to the SOL is of great interest because this can increase the SOL width, hence lowering the heat load to the divertor [5]. Handling divertor heat loads is crucial for the safe operation of plasma-facing components in future devices like ITER.

Multimachine scalings and neoclassical theory predict a very narrow SOL heat flux decay length λ_q of ~ 1 mm for ITER H-mode [6, 7], which will result in exceeding the heat load limits of plasma-facing materials in the divertor. The neoclassical heuristic drift model (HD model) for λ_q in H-mode shows quantitative agreement with the experimental data and reproduces the inverse dependence of λ_q on plasma current [6, 8]. The HD model estimates the SOL width by multiplying a typical ion residence time in the SOL, $L_{\parallel}/v_{\parallel}$, by the grad- B and curvature drift velocity perpendicular to the separatrix, which results in $\lambda_q \sim 2a\rho_p/R$, where ρ_p is the poloidal Larmor radius, and a and R are minor and major radii, respectively. This model does not account for turbulence and can be considered as a ‘neoclassical limit’ for λ_q . Numerical simulations with the gyrokinetic code XGC1 [9] and fluid turbulence modelling framework BOUT++ [10, 11] predict λ_q broadening up to 5–6 mm in ITER due to edge turbulence. A positive correlation between edge turbulence intensity and increased λ_q was observed in QH-mode plasmas at the DIII-D tokamak [12, 13]. Simulations with BOUT-HESEL code, validated on the EAST H-mode discharges, also predict a broader $\lambda_q \approx 9.3$ mm for ITER due to the proposed positive scaling of λ_q on the machine size [14]. The exact mechanism of SOL turbulence excitation and widening of λ_q is still under study and turbulence spreading can play a key role in this process [5]. Recent BOUT++ simulations for EAST discharges with small edge localized modes (ELMs) [11] and for DIII-D QH-mode [15] have shown the increase of λ_q with the increase in turbulence intensity flux $\tilde{v}_r \tilde{p}^2$ across the separatrix, which characterizes spreading. Here, $\tilde{v}_r$ is the fluctuating radial velocity and $\tilde{p}$ is the fluctuating pressure. In [11], the study of MHD turbulence in H-mode edge plasma employed a three-field model, which includes a vorticity equation, Ohm’s law, and a pressure advection equation. Thus, $\tilde{v}_r \tilde{p}^2$ is the natural measure of spreading within this model. In the case of drift wave turbulence (e.g. in L-mode), the $\langle \tilde{v}_r \tilde{n}^2 \rangle$ is an appropriate measure of spreading. It is often used for experimental estimations of turbulence intensity flux [16]. For example, the rate of turbulence spreading $\omega_s = -\frac{\partial}{\partial r} \langle \tilde{v}_r \tilde{n}^2 \rangle / \langle \tilde{n}^2 \rangle$ was measured in the SOL using Langmuir probes at the TJ-II stellarator [17]. The experiments with ohmic plasmas at the HL-2A tokamak have shown a positive

dependence of λ_q on the radial flux of turbulence internal energy $c_s^2 \langle \tilde{v}_r (\tilde{n}_e/n_e)^2 \rangle$ measured at the last closed flux surface with Langmuir probes [18, 19], with c_s denoting the ion sound speed. Turbulence spreading was determined as a dominant mechanism of SOL turbulence excitation and λ_q broadening in the case of weak edge $E \times B$ shear [19], which is consistent with recent work showing that the critical condition for SOL broadening is determined by the relation between the $E \times B$ damping and spreading from the plasma edge [5].

Turbulence spreading has been extensively studied by theory [20–25], but the physical mechanism of the spreading has not been fully experimentally characterized. Propagating pulses/avalanches are a natural candidate for a spreading mechanism. Recent results from the MAST show that correlated large-scale density fluctuations, ‘blobs’ and ‘voids’, are generated inside the separatrix with ‘blobs’ moving outward and ‘voids’ moving inward [26]. It has been shown that blobs play an important role in edge cross-field transport [27–29].

For the region outside the separatrix, an appropriate turbulence model is the Hasegawa–Wakatani system with curvature [30, 31]. Based on this model, the equation for the evolution of the turbulence intensity in the SOL is

$$\frac{1}{2} \frac{\partial \langle \tilde{n}^2 \rangle}{\partial t} = -\frac{1}{2} \frac{\partial}{\partial r} \langle \tilde{v}_r \tilde{n}^2 \rangle - \langle \tilde{v}_r \tilde{n} \rangle \frac{\partial n_0}{\partial r} - \frac{2c_s}{L_{\parallel}} \langle \tilde{n}^2 \rangle. \quad (1)$$

Here, $n = n_0 + \tilde{n}$, $v_r = v_{r0} + \tilde{v}_r$, $\langle \rangle$ denotes the zonal and time average, $\langle v_{r0} \rangle = 0$, $L_{\parallel}$ is the parallel connection length in the SOL, and c_s is the sound speed. A detailed derivation of equation (1) is shown in the appendix. The first term on the right-hand side of equation (1) includes turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ and describes spreading, the second one relates to the local drive, and the third one to the along-field-line losses.

We can approximate $\frac{\partial \langle \tilde{v}_r \tilde{n}^2 \rangle}{\partial r} \approx \frac{\langle \tilde{v}_r \tilde{n}^2 \rangle}{L_{\perp}}$, and then if

$$\frac{1}{2} \frac{\langle \tilde{v}_r \tilde{n}^2 \rangle}{L_{\perp}} > \frac{2c_s \langle \tilde{n}^2 \rangle}{L_{\parallel}}, \quad (2)$$

then the spreading flux would exceed along-field-line losses. Equation (2) will be satisfied if the radial scale of the spreading flux $L_{\perp}$ is much smaller than the parallel length scale in the SOL $L_{\parallel}$, namely $L_{\perp} \ll L_{\parallel}$. This is almost always the case, because $L_{\perp}$ is defined by the radial correlation length of the turbulence, which is in the order of 1–2 cm in our case, and $L_{\parallel}$ is the connection length from the outer midplane to the divertor and is in the order of several meters, $L_{\parallel} \sim qR$. In all cases, the quantity of interest for density turbulence spreading is the intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$.

In this work we study statistical properties (skewness and kurtosis) of density fluctuations as well as turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ at the edge of L-mode plasmas to characterize turbulence spreading and investigate the effect of spreading on the SOL width λ_q at the DIII-D tokamak. The paper is organized as follows. First, we describe a set of diagnostics used to measure the plasma parameters under study, next we show

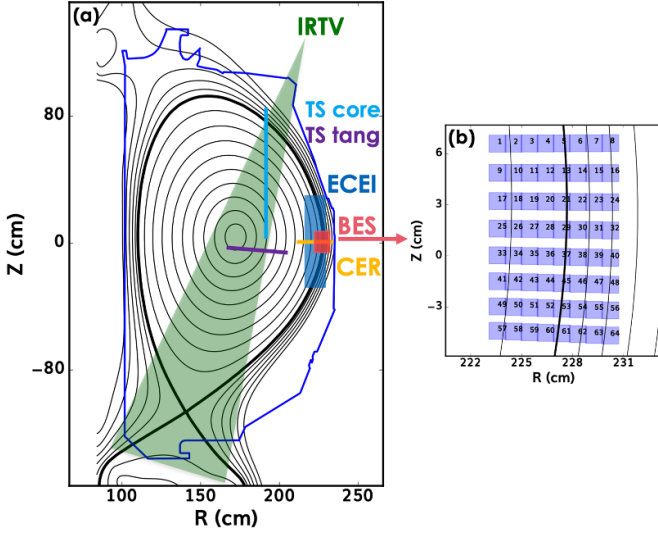


Figure 1. (a) Diagnostics layout at the DIII-D tokamak and (b) beam emission spectroscopy (BES) channel layout at the outboard midplane.

spectral analysis and radial profiles of plasma density fluctuations $\delta n/n$ measured with BES, followed by radial profiles of $\delta n/n$ skewness and kurtosis. After that, velocimetry results and calculated turbulence intensity flux profiles are presented. Finally, we discuss the observed correlation between co- I_p torque, outward turbulence spreading, and λ_q broadening.

2. Experimental setup

The experiments were performed at the DIII-D tokamak [32]. The layout of main plasma diagnostics, used for these experiments is shown in figure 1.

Long-wavelength ($k\rho_i < 1$) density fluctuations are measured using beam emission spectroscopy (BES) [33, 34]. BES collects Doppler-shifted D_α light (652–655 nm) from the heating neutral beam, passes it to high quantum efficiency photodiodes coupled with cryogenically cooled amplifiers and digitizes with a 1 MHz sampling frequency allowing turbulence measurements up to the Nyquist frequency of 500 kHz. The beam emission intensity fluctuations are proportional to plasma density fluctuations $\frac{\delta I}{I} \sim K(T_e, n_e, Z_{\text{eff}}) \frac{\delta n}{n}$ [35]. Sixty-four BES channels arranged in an 8×8 grid cover an area of ~ 10 cm poloidally and 8 cm radially at the outboard midplane, as shown in figure 1. Turbulence velocity fields are obtained from BES images using velocimetry analysis based on an orthogonal dynamic programming (ODP) algorithm [36–39]. The velocimetry processes subsequent images of BES intensity fluctuations searching for local displacements and provides quantitative measurement of a two-dimensional flow-field in the lab frame.

Electron density n_e and electron temperature T_e are measured using a Thomson scattering diagnostic [40]. The electron temperature fluctuations are measured with electron cyclotron emission imaging (ECEI) with a temporal resolution of $1 \mu\text{s}$ [41]. The ECEI spot size is 3 cm poloidally and 1.5 cm radially,

which limits the measured wavenumbers to $k_{\text{pol}} \leq 1 \text{ cm}^{-1}$ and $k_r \leq 2 \text{ cm}^{-1}$ [42]. The divertor heat flux profile at the outer strike point is measured using an infrared television camera, IRTV [43]. The radial electric field E_r is calculated from the radial force balance equation for fully stripped carbon ions [44]. Density, temperature, and the velocities of poloidal and toroidal rotation V_{pol} and V_{tor} of carbon ions are measured with a charge exchange recombination spectroscopy (CER) system [45, 46].

3. Experimental results

3.1. Turbulence properties in ECH and NBI power ramps and torque scan

Turbulence characteristics were investigated at the DIII-D tokamak in L-mode deuterium plasmas in three scenarios: ECH power ramp ($P_{\text{ECH}} = 0.25, 0.5, 1.5 \text{ MW}$), NBI power ramp ($P_{\text{NBI}} = 1.2, 1.7, 2.5 \text{ MW}$) and injected torque scan ($T_{\text{inj}} = -1, -0.4, 0.2, 0.6 \text{ Nm}$) where a positive injected torque is in the co-current direction. The toroidal magnetic field was $B_t = 2 \text{ T}$ and the plasma current $I_p = 1 \text{ MA}$. All the discharges have B_t in a ‘reversed’ direction (counter-clockwise when viewing from the top of the machine), which ensures a higher L–H power threshold with ion ∇B drift directed away from the X-point in a lower single null configuration. I_p is also ‘reversed’ (clockwise when viewing from the top of the machine) to ensure a proper magnetic field line pitch angle alignment for BES measurements. The time traces of the main plasma parameters are shown in figure 2, and mean values of the parameters are collected in table 1.

Edge electron density n_e , electron temperature T_e , and electron pressure gradient profiles $\frac{dp_e}{d\rho}$ for ECH and NBI power ramps are shown in figure 3; n_e , T_e , $\frac{dp_e}{d\rho}$, and carbon toroidal rotation frequency ω_{tor} profiles for the torque scan are shown in figure 4. Radial profiles of the edge radial electric field E_r are shown in figure 5. E_r profiles are reconstructed from radial force balance using the pressure gradient and V_{pol} and V_{tor} measured with impurity (carbon) CER. The electron pressure gradient becomes significantly stronger only at the highest power level for both ECH and NBI power ramps, with the maximum amplitude around $\rho \sim 0.97$ – 0.98 . During the torque scan, $\frac{dp_e}{d\rho}$ does not change within the error bars. The E_r well becomes deeper only at the highest power level as well, with the minimum value at the same location as the maximum $|\frac{dp_e}{d\rho}|$ ($\rho \sim 0.97$). During the torque scan, the E_r profile is mostly flat and does not show significant changes within the error bars.

Spectra of cross-power G , cross-phase θ , and coherence γ^2 between two poloidally separated BES channels measured at $\rho \sim 0.97$ at different ECH/NBI power levels are shown in figure 6. Here we use standard definitions for G , θ , and γ^2 [47]:

$$G_{12}(f) = F^*(I_1)F(I_2) \quad (3)$$

$$\theta_{12}(f) = \arg(G_{12}(f)) \quad (4)$$

$$\gamma_{12}^2 = \frac{|G_{12}(f)|^2}{G_{11}(f)G_{22}(f)} \quad (5)$$

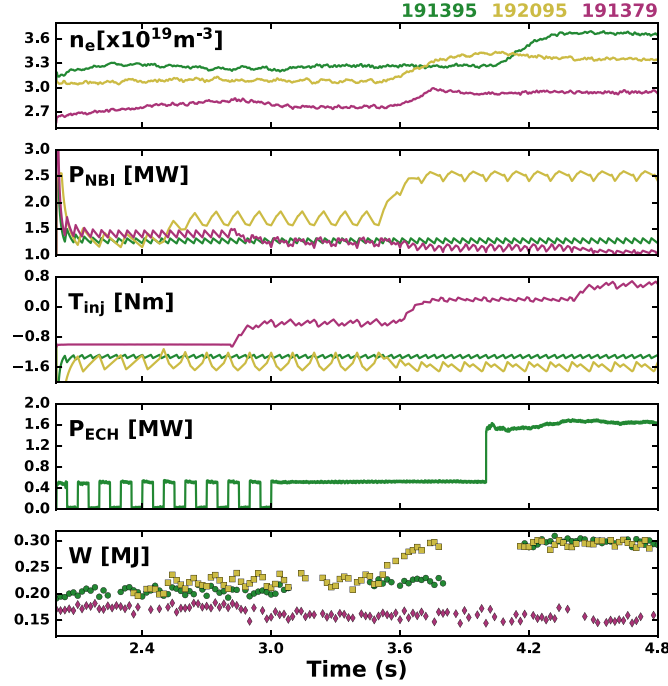


Figure 2. Time evolution of line-averaged density (a), injected NBI power (b), injected torque (positive T_{inj} is co- I_p) (c), injected ECH power (d), and plasma stored energy (e). ECH power ramp is in green, NBI power ramp is in yellow, and torque scan is in purple.

Table 1. Line-averaged electron density, injected NBI power, injected ECH power, and injected torque for the shots under study. All shots had $B_t = 2$ T and $I_p = 1$ MA.

Shot #/Scenario	n_e (10^{19} m^{-3})	P_{NBI} (MW)	P_{ECH} (MW)	T_{inj} (Nm)
#191395 ECH power ramp	3.3–3.7	1.2	0.25, 0.5, 1.5	–1.5
#192095 NBI power ramp	3.1–3.4	1.2, 1.7, 2.5	0	–1.5
#191379 Torque scan	2.75–2.9	1.2	0	–1, –0.4, 0.2, 0.6

where I_1 and I_2 are measured BES signals, and F and F^* denote finite Fourier transform and its complex conjugate.

Each spectrum is calculated over a 500 ms time slice in the ECH power ramp and a 400 ms time slice in the NBI power ramp discharge. The poloidal distance between the channels is $\Delta_{pol} \sim 1$ cm. High coherence is observed for both ECH and NBI power ramps in the broad frequency band for $f < 130$ kHz. Both cross-power and coherence increase with injected power. A positive cross-phase spectrum with a linear dependence on frequency indicates rotation to the electron diamagnetic drift direction (EDD). A strong coherent peak is observed at $f \sim 5$ kHz; it most probably has an MHD nature, and its properties are beyond the scope of this paper.

The amplitude of beam emission intensity fluctuations $\delta I/I$ is calculated using the cross-power spectrum as follows:

$$\frac{\delta I}{I} = \sqrt{\frac{\left(\int_{f_1}^{f_2} G_{12}(f) df - W_{\text{photon}} \right)}{\bar{I}_1 \bar{I}_2}} \quad (6)$$

where the integral is taken over the frequency band $f = 7$ – 130 kHz, $\bar{I}_1$ and $\bar{I}_2$ are time-averaged amplitudes of each of the BES signals, and W_{photon} is the equivalent energy of the photon noise. The power of the photon noise is assumed constant over the whole frequency band and is estimated as an average cross-power in the frequency band $f = 200$ – 250 kHz as we do not observe any coherent structures in the spectra at $f > 160$ kHz. The amplitude of density fluctuations is calculated using the aforementioned relation $\frac{\delta I}{I} \sim K(T_e, n_e, Z_{\text{eff}}) \frac{\delta n}{n}$ using n_e and T_e profiles; the assumption of $Z_{\text{eff}} = 2$ is used for all radial locations.

Spectra of density fluctuations measured with BES agree well with the spectra of ECE radiation intensity, or radiation

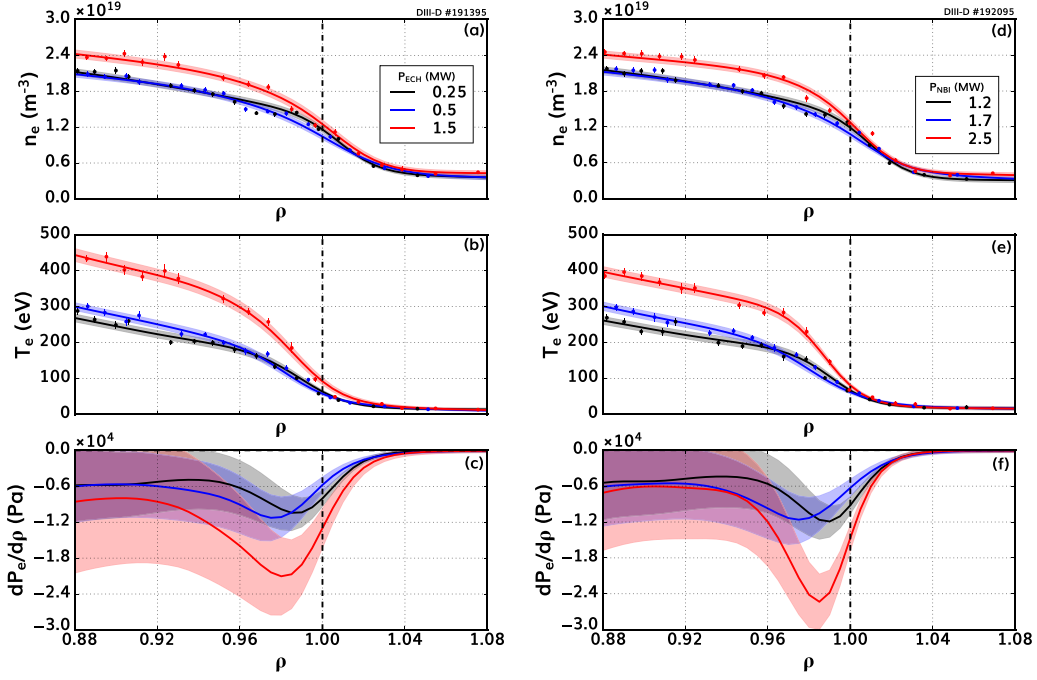


Figure 3. Radial profiles of edge electron density n_e (a), electron temperature T_e (b), and electron pressure gradient $\frac{dP_e}{d\rho}$ (c) for the ECH power ramp, shot #191395, $P_{ECH} = 0.25$ MW in black, 0.5 MW in blue, 1.5 MW in red; n_e (d), T_e (e), and $\frac{dP_e}{d\rho}$ (f) profiles for the NBI power ramp, shot #192095, $P_{NBI} = 1.2$ MW in black, 1.7 MW in blue, and 2.5 MW in red. Dashed lines show the location of the separatrix.

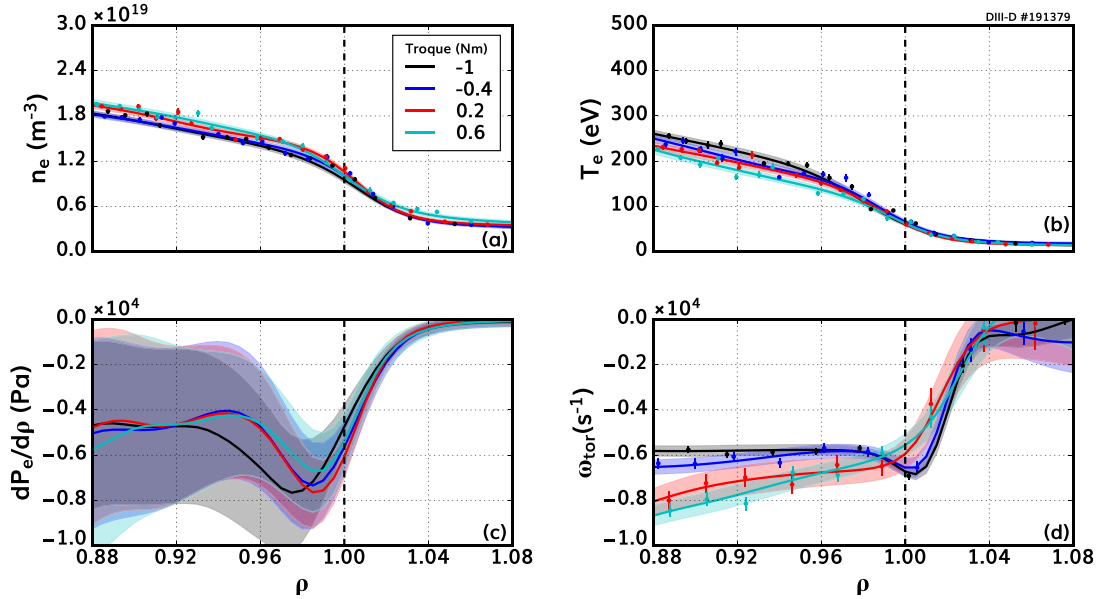


Figure 4. Radial profiles of edge electron density n_e (a), electron temperature T_e (b), electron pressure gradient $\frac{dP_e}{d\rho}$ (c), and carbon ion toroidal rotation frequency (d) for the torque scan, shot #191379, $T_{inj} = -1$ Nm in black, -0.4 Nm in blue, 0.2 Nm in red, and 0.6 Nm in cyan. $T_{inj} > 0$ is co- I_p . Dashed line shows the location of the separatrix.

temperature, T_{e_rad} measured with ECEI, as seen from figure 7. Figure 7(a) shows the spectrum of coherence and figure 7(b) shows the poloidal wavenumber spectrum for two poloidally separated BES (in blue) and ECEI (in green) channels at $\rho \sim 0.97$. The measurements are taken during the NBI power ramp with $P_{NBI} = 1.7$ MW. The poloidal wavenumber spectrum was obtained by dividing the cross-phase spectrum by

the poloidal distance between channels. For ECEI this distance is approximately 3 cm, which is ~ 2 times larger than that for BES, so we chose a pair of BES channels with a similar 3 cm separation. High coherence of T_{e_rad} fluctuations is observed in a narrower frequency band ($f = 20\text{--}80$ kHz), which can be explained by the differences in respective point-spread-functions and corresponding wavenumber sensitivity,

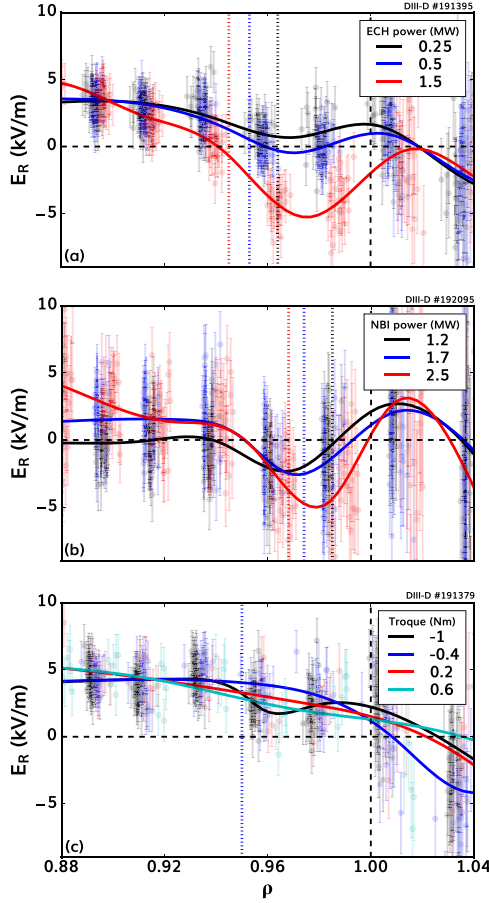


Figure 5. Radial profiles of edge radial electric field E_r in ECH power ramp (a), NBI power ramp (b), and torque scan (c). Colored dots with error bars show the E_r values obtained from CER data. Colored dotted vertical lines show the locations of zero $\delta n/n$ skewness.

with lower ECEI sensitivity to small-scale fluctuations compared to BES [42]. The poloidal wavenumber spectra in both diagnostics show similar values of $k_{\text{pol}} \sim 0.6 \text{ rad cm}^{-1}$. It should be noted that the measured radiation temperature $T_{\text{e,rad}}$ is solely determined by the electron temperature T_e at the ECE cold resonance location only in the approximation of black-body radiation. This approximation is valid if plasma is optically thick for ECE radiation, with high values of the optical depth $\tau > 3$ ($\tau \propto n_e, T_e$) [48]. At $\rho \sim 0.97$ where the ECEI turbulence spectrum measurement is made, the ECE radiation is in an optical grey regime with $\tau \sim 1$, as seen from figure 8 showing the edge radial profiles of the optical depth, electron density, and electron temperature. A non-local ECE radiation (shine-through) effect emerges and the measurement at $\rho \sim 0.97$ can be a result of turbulence fluctuation of hot electrons at $\rho < 0.97$. However, in the L-mode plasma edge where $T_e \leq 300 \text{ eV}$, the non-local ECE radiation effect is weak [48–50] and the $\delta T_{\text{e,rad}}$ can be interpreted as local mixed δn_e and δT_e . The next inner ECEI channel at $\rho \sim 0.92$ cannot see or sees a very weak fluctuation, which also supports that the ECEI spectrum at $\rho \sim 0.97$ is from local edge turbulence fluctuation at $\rho \geq 0.92$. In addition, the ECE sensitivity to both δn_e and δT_e due to the optical grey regime does not affect the measurement

of the turbulence poloidal wavenumber [48]. In the SOL, the ECEI measurements are not localized, because τ is well below 1, meaning low absorption at the resonant layer, which leads to a shine-through of the emission from different plasma layers [51, 52].

Radial profiles of normalized density fluctuation amplitudes $\delta n/n$ for the frequency band $f = 7\text{--}130 \text{ kHz}$ during ECH and NBI power ramps are shown in figure 9. Dashed lines show the radiation temperature $\delta T_{\text{e,rad}}/T_{\text{e,rad}}$ fluctuation amplitude for the same frequency band. Two radial locations are plotted because only these channels show the EDD mode and are located in the region with sufficient optical depth. Profiles were calculated over a 500 ms time slice for the ECH and a 400 ms time slice for NBI power ramps. The amplitude of $\delta n/n$ peaks at $\rho \sim 0.94$ in the ECH power ramp discharge and at $\rho \sim 1$ in the NBI power ramp discharge. $\delta n/n$ increases by 40% with the increase of ECH power from $\sim 2\%$ to $\sim 3\%$. It should be noted that in NBI power ramp only $\delta n/n$ ($\rho = 0.97$) increases and $\delta n/n$ ($\rho = 1$) remains roughly constant. $\delta T_{\text{e,rad}}/T_{\text{e,rad}}$ at $\rho \sim 0.98$ increases with ECH and NBI power from 1.6% to 3.5% and from 1% to 2.5%, respectively.

Spectra of cross-power, cross-phase, and coherence during the torque scan are shown in figure 10. Each spectrum is calculated over a 300 ms time slice. In the torque scan, the cross-phase spectra show a double mode structure (see figure 10) with lower frequencies $f \sim 5\text{--}20 \text{ kHz}$ rotating in the EDD (positive cross-phase) and higher frequencies $f \sim 20\text{--}130 \text{ kHz}$ rotating in the ion diamagnetic drift direction (IDD, negative cross-phase). From cross-power and coherence spectra it is hard to make a clear distinction between the two modes, since they overlap in frequency, so we calculate the amplitude of density fluctuations $\delta n/n$ by integrating the cross-power over the frequency band $f = 5\text{--}130 \text{ kHz}$ that covers both modes.

Radial profiles of $\delta n/n$ for different values of injected torque are shown in figure 11. Profiles were calculated over a 300 ms time slice. When the torque changes from counter- I_p ($T_{\text{inj}} < 0$) to co- I_p ($T_{\text{inj}} > 0$), the coherence in this frequency band becomes stronger, as shown in figure 10(c), and the fluctuation amplitude $\delta n/n$ increases from 3.1% to $\sim 4.7\%$. $\delta T_{\text{e,rad}}/T_{\text{e,rad}}$ is shown in figure 11 for two radial locations; the maximum amplitude is $\sim 1.2\%$ at $\rho \sim 0.98$, and it does not change with torque.

3.2. Statistics of density fluctuations

Skewness and kurtosis of BES signals are calculated to characterize the probability density function (PDF) of density fluctuations in all three scenarios. For a sample distribution x ; skewness is the third and kurtosis is the fourth moment, which are calculated as follows:

$$m_j = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^j \quad (7)$$

$$\text{Skewness} = \frac{m_3}{m_2^{3/2}} \quad (8)$$

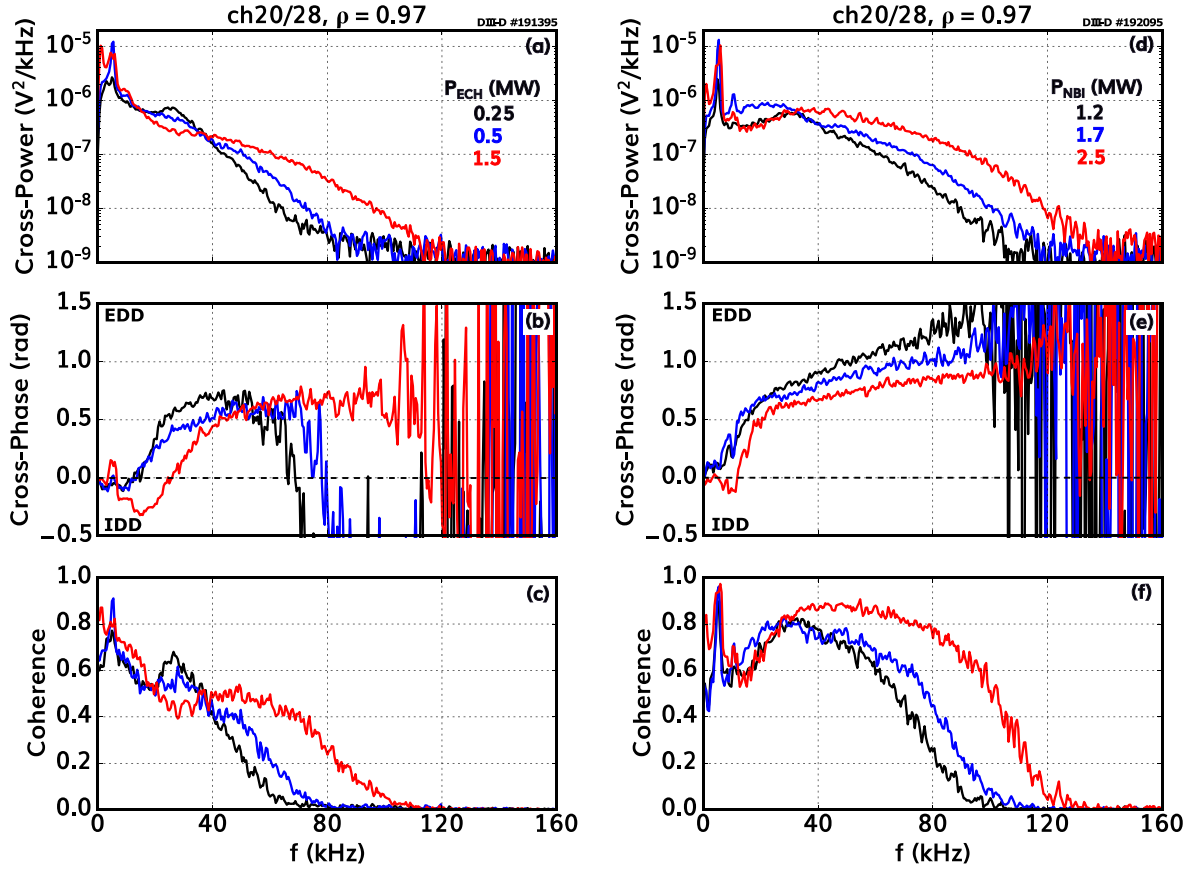


Figure 6. Cross-power (a), cross-phase (b), and coherence (c) spectra for two poloidally separated BES channels at $\rho = 0.97$, $\Delta_{\text{pol}} \sim 1$ cm in ECH power ramp, shot #191395. $P_{\text{ECH}} = 0.25$ MW in black, 0.5 MW in blue, and 1.5 MW in red. Spectra are calculated over a 500 ms time slice. Cross-power (d), cross-phase (e), and coherence (f) spectra between two poloidally separated channels at $\rho = 0.97$, $\Delta_{\text{pol}} \sim 1$ cm in NBI power ramp, shot #192095. $P_{\text{NBI}} = 1.2$ MW in black, 1.7 MW in blue, and 2.5 MW in red. Spectra are calculated over a 400 ms time slice.

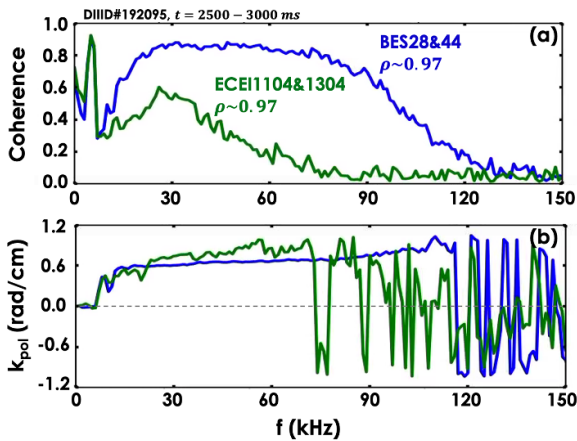


Figure 7. Coherence (a) and poloidal wavenumber (b) spectra for two poloidally separated BES channels (blue) and ECEI channels (green), $P_{\text{NBI}} = 1.7$ MW, shot #192095. The poloidal channel separation for both diagnostics is 3 cm.

$$\text{Kurtosis} = \frac{m_4}{m_2^2} - 3. \quad (9)$$

The sign of skewness shows the shift of the PDF to negative or positive values and kurtosis follows the shape of the

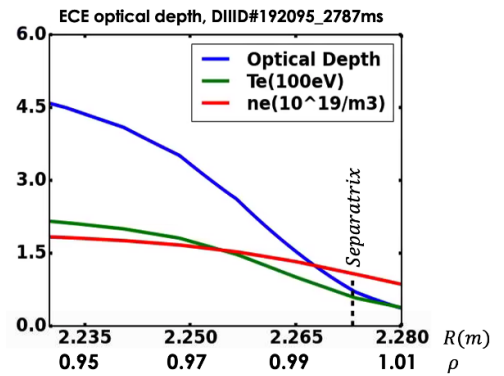


Figure 8. Radial profiles of the ECE optical depth (blue), electron temperature (green), and electron density (red).

distribution: it is zero for a normal distribution, negative for a distribution closer to uniform, and positive for a distribution with ‘heavy’ tails. An example of synthetic signals with different PDFs and different values of skewness and kurtosis is shown in figure 12. The blue signal has a normal distribution so both skewness and kurtosis are zero. The orange signal has a PDF with heavy tails, which indicates a predominance of large

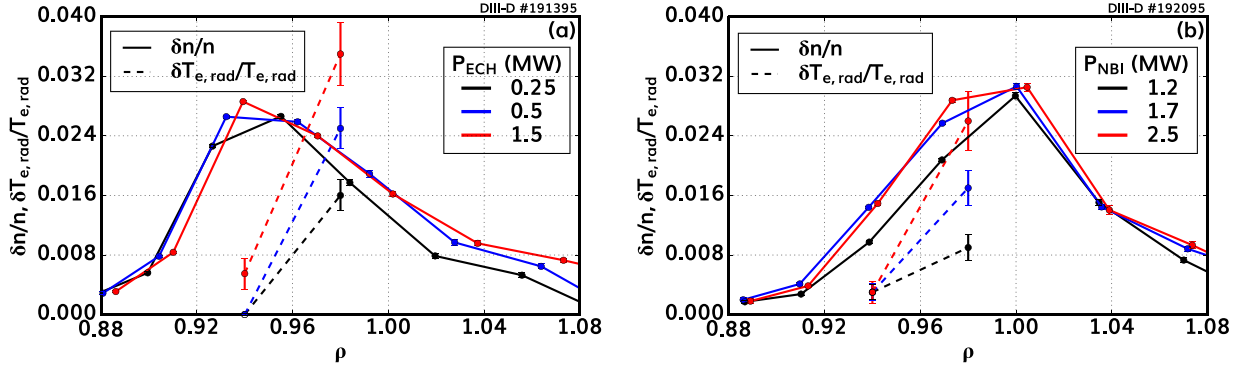


Figure 9. Radial profiles of density and radiation temperature fluctuation amplitude for ECH (a) and NBI (b) power ramps in the frequency band $f = 7\text{--}130$ kHz. Profiles are calculated over 500 ms and 400 ms time slices, respectively.

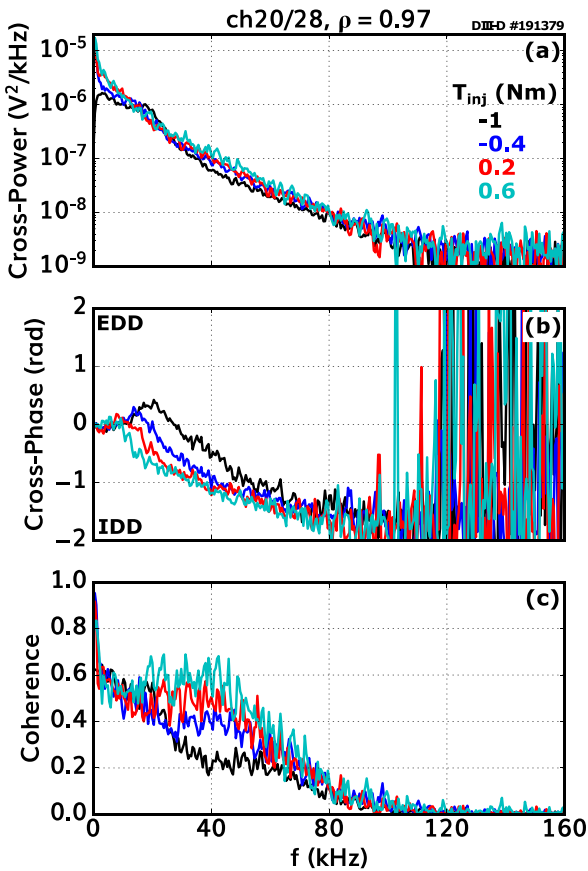


Figure 10. Cross-power (a), cross-phase (b), and coherence (c) spectra for two poloidally separated BES channels at $\rho = 0.97$, $\Delta_{\text{pol}} \sim 1$ cm in torque scan, shot #191379. $T_{\text{inj}} = -1$ Nm in black, -0.4 Nm in blue, 0.2 Nm in red, and 0.6 Nm in cyan. $T_{\text{inj}} > 0$ is co- I_p . Spectra are calculated over a 300 ms time slice.

amplitude fluctuations, and it is reflected in a positive kurtosis. The magenta signal has more positive fluctuations, so its PDF is shifted to positive values with a heavy positive tail and both skewness and kurtosis are positive.

Skewness and kurtosis are calculated for all 64 BES channels and averaged over the poloidal direction. Raw BES signals are frequency-filtered to isolate coherent turbulence: $f = 20\text{--}120$ kHz for power ramps and $f = 10\text{--}120$ kHz for

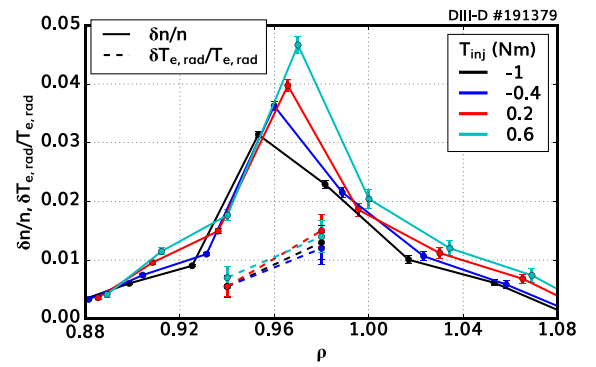


Figure 11. Radial profiles of density and radiation temperature fluctuation amplitude during torque scan for the frequency band $f = 5\text{--}130$ kHz, shot #191379. $T_{\text{inj}} = -1$ Nm in black, -0.4 Nm in blue, 0.2 Nm in red, and 0.6 Nm in cyan. $T_{\text{inj}} > 0$ is co- I_p . Profiles are calculated over a 300 ms time slice.

the torque scan. Radial profiles of skewness and kurtosis of $\delta n/n$ for the ECH and NBI power ramps, and the torque scan are shown in figures 13–15, respectively. Nonzero skewness and kurtosis values observed across the radial range $\rho = 0.88\text{--}1.04$ indicate that turbulent density fluctuations have a non-Gaussian distribution. In all cases, the skewness of $\delta n/n$ changes its sign around $\rho = 0.93\text{--}0.98$; it is negative at inner radii demonstrating that there are more negative fluctuations or ‘voids’, and it is positive at outer radii, which means there are more positive fluctuations or ‘blobs’. This result agrees with earlier $\delta n/n$ skewness measurements at the DIII-D [53], JET [54], ASDEX Upgrade [55], HL-2A [56], EAST [57, 58], and MAST [26]. It should be noted that previous measurements at the DIII-D tokamak [53] showed zero crossing of $\delta n/n$ skewness at the separatrix: the inconsistency may be caused by uncertainties in the determination of the separatrix location. Simulations with XGC1 for DIII-D H-mode [59] and BOUT++ simulations for DIII-D QH-mode [15] show a similar radial profile of $\delta n/n$ skewness as well.

During ECH and NBI power ramps, the skewness of $\delta n/n$ becomes more negative at inner radii and more positive at outer radii, as shown in figures 13(a) and 14(a). The location of zero skewness, which can be considered as the birthplace of

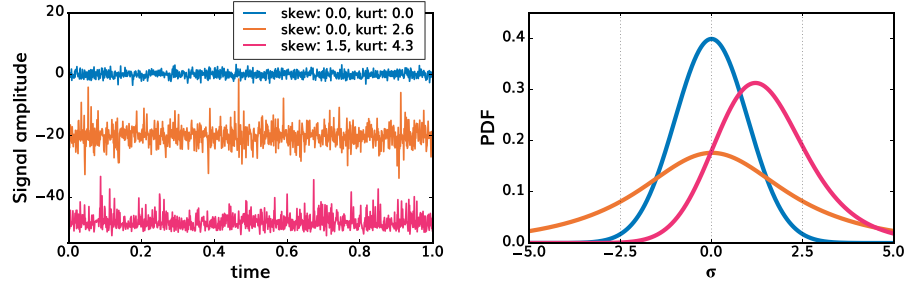


Figure 12. Example of skewness and kurtosis calculated for synthetic signals with different PDFs.

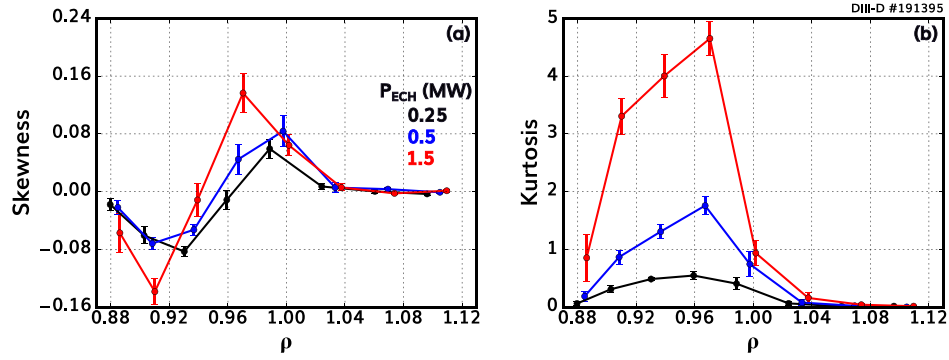


Figure 13. Radial profiles of skewness (a) and kurtosis (b) of density fluctuations in the frequency band $f = 20\text{--}120$ kHz, ECH power ramp, shot #191395. $P_{\text{ECH}} = 0.25$ MW in black, 0.5 MW in blue, and 1.5 MW in red.

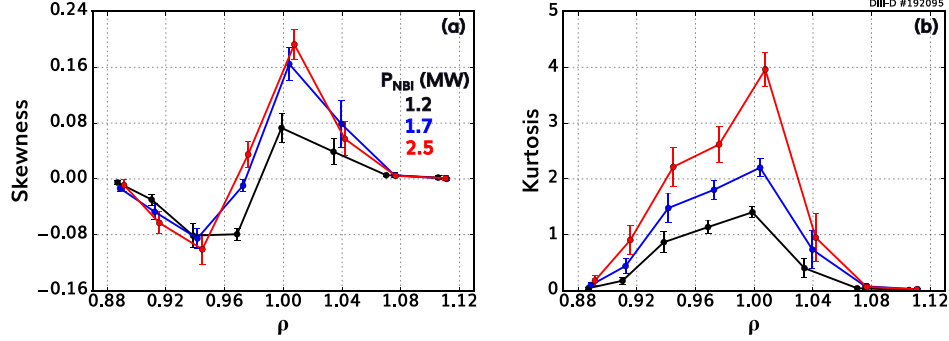


Figure 14. Radial profiles of skewness (a) and kurtosis (b) of density fluctuations in the frequency band $f = 20\text{--}120$ kHz, NBI power ramp, shot #192095, $P_{\text{NBI}} = 1.2$ MW in black, 1.7 MW in blue, and 2.5 MW in red.

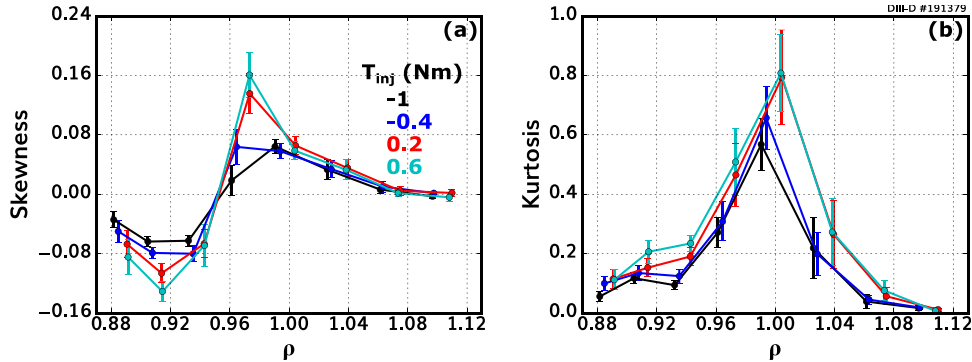


Figure 15. Radial profiles of skewness (a) and kurtosis (b) of density fluctuations in the frequency band $f = 10\text{--}120$ kHz, torque scan, shot #191379. $T_{\text{inj}} = -1$ Nm in black, -0.4 Nm in blue, 0.2 Nm in red, and 0.6 Nm in cyan. $T_{\text{inj}} > 0$ is co- I_p .

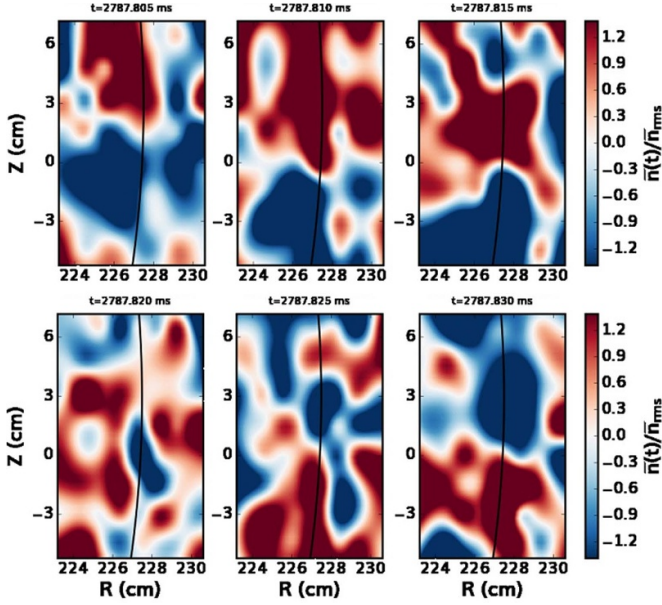


Figure 16. BES turbulence images at $5 \mu\text{s}$ time intervals (left to right), $P_{\text{NBI}} = 1.7 \text{ MW}$, shot #192095. BES signals are filtered to $f = 20\text{--}120 \text{ kHz}$ and normalized by the standard deviation. The black line shows the separatrix. The color scale shows the level of density fluctuations normalized by its root-mean-square (rms) value.

‘blob–void’ pairs, is located in the area of stronger E_r shear and moves inward with the increase of ECH/NBI power. The location of zero skewness is marked by colored vertical dotted lines in figure 5. During the torque scan, the skewness of $\delta n/n$ decreases gradually at $\rho < 0.95$, but at $\rho > 0.95$ it increases only when the injected torque changes direction from counter- I_p to co- I_p , see figure 15(a). The location of zero skewness does not move during the torque scan. In all cases, the zero-skewness point is located deeper inside than the maximum of $\left| \frac{dP_e}{d\rho} \right|$. The kurtosis of $\delta n/n$ increases significantly from 0.5–1 to 4–5 with injected ECH/NBI power over a wide radial range $\rho = 0.88\text{--}1.04$, which means that there are more large $\delta n/n$ fluctuation amplitudes at higher power; see figures 13(b) and 14(b). During the torque scan, the kurtosis of $\delta n/n$ only increases from 0.6 to 0.8 when T_{inj} changes from counter- I_p to co- I_p , as shown in figure 15(b).

3.3. Turbulence velocities and turbulent intensity flux measurements

A velocimetry ODP algorithm is used to retrieve the velocity field from a sequence of BES turbulence images [37, 39]. The 8×8 resolution of raw images is too low to be directly used by the ODP algorithm, so the images are spatially interpolated to a higher resolution of 40×40 using a radial basis function interpolation with a cubic norm. As an example, a set of images created using frequency-filtered and normalized BES signals at $5 \mu\text{s}$ time intervals is shown in figure 16. The images show turbulent eddies moving poloidally down, which is the direction of electron diamagnetic drift. A similar picture is observed for radiation temperature fluctuations $\delta T_{e,\text{rad}}/T_{e,\text{rad}}$ as measured

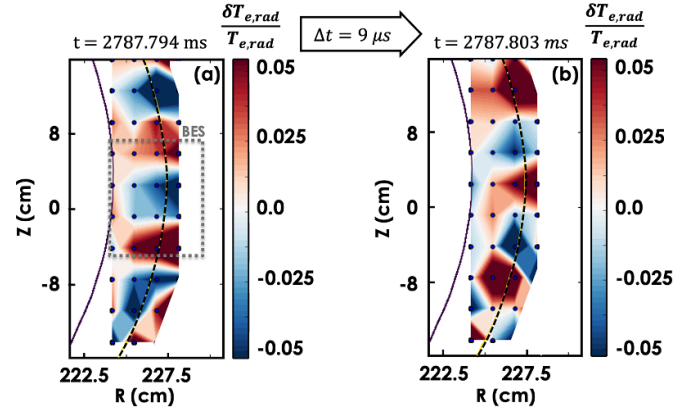


Figure 17. ECEI turbulence images at the outboard midplane with $9 \mu\text{s}$ time intervals, $f = 20\text{--}120 \text{ kHz}$, $P_{\text{NBI}} = 1.7 \text{ MW}$, shot #192095. The black dashed line shows the separatrix. The grey dashed rectangle represents the area of BES measurements.

by ECEI. Figure 17 shows two images of $\delta T_{e,\text{rad}}/T_{e,\text{rad}}$ in the frequency band $f = 20\text{--}120 \text{ kHz}$. Images are taken at $9 \mu\text{s}$ time intervals and show that temperature fluctuations move poloidally down (in the EDD).

Velocity fields are calculated for 20 ms time slices at each ECH/NBI power and torque level and then averaged over time and over the poloidal direction to obtain radial profiles. The innermost and the outermost points were excluded due to larger errors in the output of the ODP algorithm, which can occur at the boundary of the image. Figures 18–20 show radial profiles of time-averaged radial velocity $\langle v_r \rangle$ (panel (a)), poloidal velocity $\langle v_{\text{pol}} \rangle$ (panel (b)), turbulent particle flux $\langle \tilde{v}_r \tilde{n} \rangle$ (panel (c)), and turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ (panel (d)) for ECH power ramp, NBI power ramp, and torque scan scenarios, respectively. In all cases, we use $\tilde{v}_r = v_r - \langle v_r \rangle$ and $\tilde{n} \equiv \delta n/n$ to take into account turbulence amplitude, where $\langle \rangle$ denote the time averaging. Positive values of v_r mean that it is directed outward and negative v_r is inward (the same for calculated fluxes). Positive values of v_{pol} denote the rotation to the IDD and negative values denote the rotation to the EDD.

Figures 18 and 19 show that the dynamics of the calculated quantities is similar during ECH and NBI power ramps: the time-averaged v_r profile (panel (a) in figures 18 and 19) does not change with power and is negative inside the separatrix where the skewness of $\delta n/n$ is negative, indicating that negative density fluctuations, ‘voids’, move inward. In the ECH power ramp scenario, v_r changes sign from negative to positive at the separatrix; in NBI power ramp, v_r becomes positive right outside the separatrix at $\rho \sim 1.03$; in both cases, the skewness of $\delta n/n$ is positive at these radii, indicating that positive density fluctuations, ‘blobs’, move outward. During power ramps, the skewness of $\delta n/n$ zeroes out at $\rho \sim 0.88$, which is approximately 4 cm inside the separatrix; this distance can be interpreted as the depth of void propagation inside the plasma. Radial profiles of v_{pol} (panel (b) in figures 18 and 19) show the rotation to the EDD inside the separatrix. The amplitude of v_{pol} is 2–3 times larger than the amplitude of v_r .

During the ECH power ramp, changes in v_{pol} are observed at $\rho < 0.96$, see figure 18(b). The amplitude of v_{pol} at

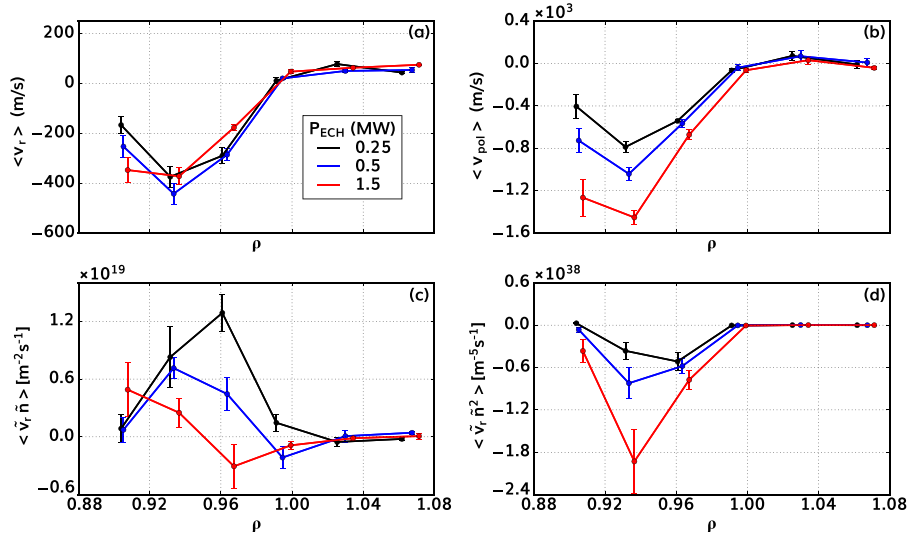


Figure 18. Radial profiles of time-averaged radial velocity $\langle v_r \rangle$ (a), poloidal velocity $\langle v_{pol} \rangle$ (b), turbulent particle flux $\langle \tilde{v}_r \tilde{n} \rangle$ (c), and turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ (d) for ECH power ramp, shot #191395. $P_{ECH} = 0.25$ MW is in black, 0.5 MW in blue, and 1.5 MW in red.

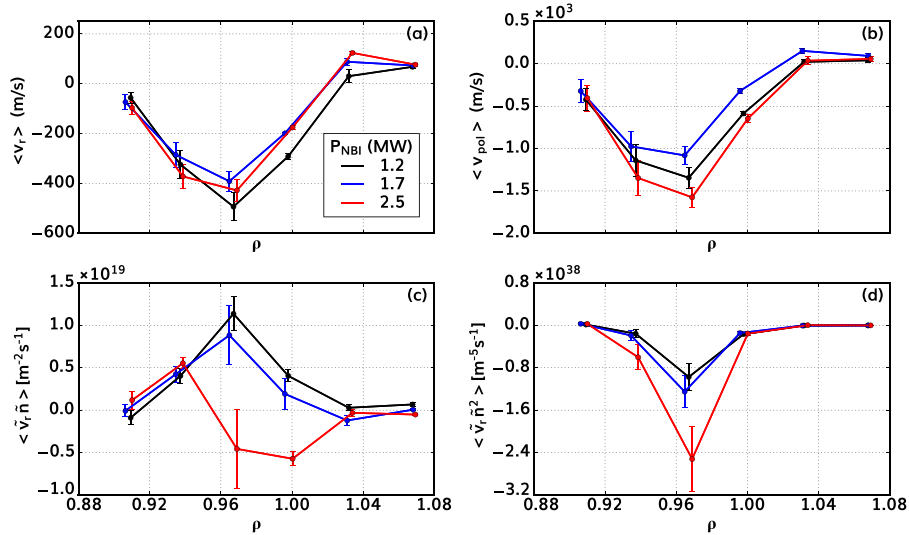


Figure 19. Radial profiles of time-averaged radial velocity $\langle v_r \rangle$ (a), poloidal velocity $\langle v_{pol} \rangle$ (b), turbulent particle flux $\langle \tilde{v}_r \tilde{n} \rangle$ (c), and turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ (d) for NBI power ramp, shot #192095. $P_{NBI} = 1.2$ MW is in black, 1.7 MW in blue, and 2.5 MW in red.

$\rho \sim 0.94$ decreases by $\sim 20\%$ when ECH power increases from 0.25 MW to 0.5 MW and then increases by $\sim 70\%$ at the highest level of ECH power (1.5 MW). During the NBI power ramp, the most pronounced changes in v_{pol} are observed at $0.94 < \rho < 1$, see figure 19(b). The amplitude of v_{pol} at $\rho \sim 0.96$ decreases by $\sim 10\%$ when NBI power increases from 1.2 MW to 1.7 MW and then increases by $\sim 50\%$ at the highest level of NBI power (2.5 MW). The increase in v_{pol} amplitude at higher power is consistent with the deeper E_r well.

During the torque scan, significant changes in both v_r and v_{pol} profiles occur at $0.92 < \rho < 1$ as shown in figures 20(a) and (b). When the injected torque changes from counter- I_p ($T_{inj} < 0$) to co- I_p ($T_{inj} > 0$), v_r amplitude decreases but stays negative (inward) at $\rho < 0.95$ where skewness of $\delta n/n$ is negative and is positive (outward) at $\rho > 1$, indicating inward movement of ‘voids’ at inner radii and outward movement of ‘blobs’

outside the separatrix. However, as the injected torque reaches its maximum co- I_p value ($T_{inj} = 0.6$ Nm) v_r changes its sign from negative to positive at $\rho \sim 0.97$, which means that ‘blobs’ start moving outward from the plasma edge to the SOL. v_{pol} changes its direction from the EDD to the IDD at $\rho < 0.96$ as T_{inj} changes from -1 Nm to -0.4 Nm and keeps increasing in amplitude as T_{inj} switches to co- I_p , see figure 20(b).

Turbulent particle flux $\langle \tilde{v}_r \tilde{n} \rangle$ is directed outward at $\rho = 0.94-1$ in most cases as seen from panel (c) in figures 18–20. Inward $\langle \tilde{v}_r \tilde{n} \rangle$ is observed at the highest level of ECH power (panel (c) in figure 18, red curve) and at the highest level of NBI power (panel (c) in figure 19, red curve): these cases require further investigation. Decreased $\langle \tilde{v}_r \tilde{n} \rangle$ is observed at low co- I_p torque, $T_{inj} = 0.2$ Nm, as shown in panel (c) figure 20, which also requires a more detailed analysis in the future.

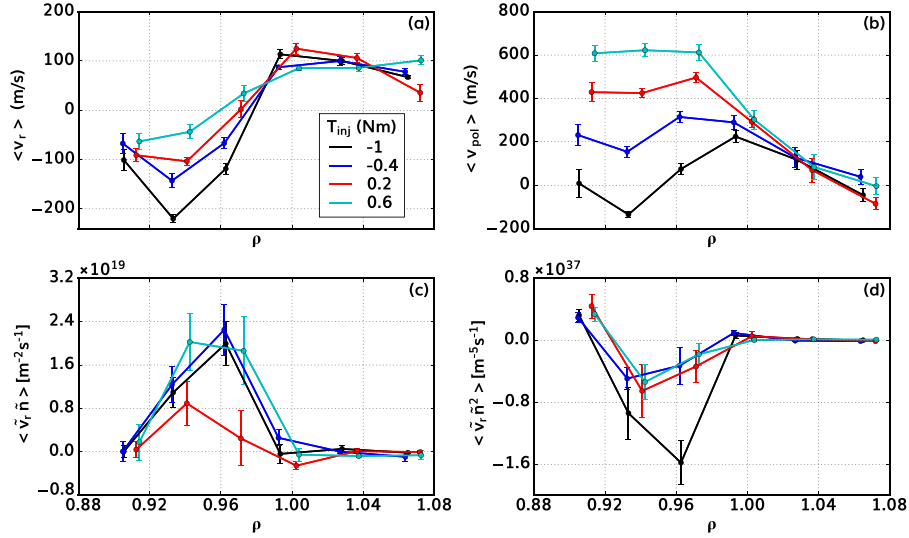


Figure 20. Radial profiles of time-averaged radial velocity $\langle v_r \rangle$ (a), poloidal velocity $\langle v_{pol} \rangle$ (b), turbulent particle flux $\langle \tilde{v}_r \tilde{n} \rangle$ (c), and turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ (d) for torque scan, shot #191379. $T_{inj} = -1$ Nm is in black, -0.4 Nm in blue, 0.2 Nm in red, and 0.6 Nm in cyan. $T_{inj} > 0$ is co- I_p .

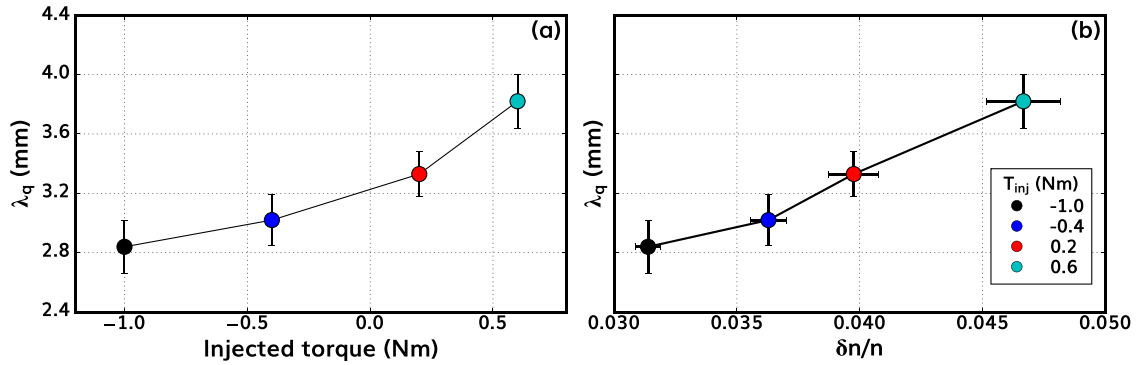


Figure 21. Trends in SOL heat flux decay length λ_q and normalized turbulence amplitude $\delta n/n$ during the torque scan, shot #191379. (a) λ_q vs injected torque T_{inj} ; (b) λ_q vs $\delta n/n$ at $\rho \sim 0.97$. Marker colors in panel (b) correspond to the same T_{inj} values as in panel (a).

The turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ is inward at $\rho = 0.92$ – 1 and zero within the error bars outside the separatrix in ECH and NBI power ramp scenarios, see panel (d) in figures 18 and 19. The amplitude of $\langle \tilde{v}_r \tilde{n}^2 \rangle$ increases with power. An inward-directed turbulence intensity flux inside the separatrix demonstrates turbulence spreading towards the core plasma, which may be responsible for turbulence excitation in the inner edge-core coupling region, the so-called ‘no man’s land’ [21, 22]. The studies of turbulence spreading to the ‘no man’s land’ require BES measurements at inner radii and are beyond the scope of this study. These observations also show the connection between inward turbulence spreading and inward movement of ‘voids’: strong inward turbulence intensity flux is observed in the radial region where skewness of $\delta n/n$ is negative.

During the torque scan, there is a significant reduction in inward $\langle \tilde{v}_r \tilde{n}^2 \rangle$ as the injected torque is switched from strong counter- I_p (-1 Nm) to weak counter- I_p (-0.4 Nm) and co- I_p (0.2 and 0.6 Nm), see panel (d) in figure 20. Notably, the absolute values of $\langle \tilde{v}_r \tilde{n}^2 \rangle$ are significantly lower than in power ramps.

The natural question is what is the contribution of turbulence spreading to SOL turbulence drive compared to local turbulence production in the SOL? To compare the impact of turbulence spreading from the separatrix with the local drive in the SOL, the production ratio $R_a = \frac{\langle \tilde{v}_r \tilde{n}^2 \rangle|_{sep}}{\int_{sep}^{SOL} -\langle \frac{\partial n}{\partial r} \rangle \langle \tilde{v}_r \tilde{n} \rangle dR}$ can be used [11, 19]. Unfortunately, the amplitude of turbulence measured with BES outside the separatrix is very weak, and calculated turbulent particle fluxes are near zero with large error bars, which makes problematic the estimation of R_a in this experiment. Dedicated experiments are needed to measure the turbulence production ratio R_a in the SOL.

3.4. Heat flux width broadening during injected torque scan

Divertor heat flux profiles are measured at the outer strike point using infrared camera data. The SOL heat flux decay length λ_q is calculated using an Eich fit [6] of the measured outer strike point heat flux. We find no trends in λ_q with ECH/NBI power, with $\lambda_q \approx 2.7$ mm in the ECH power ramp and 3.4 mm in the NBI power ramp. However, we do observe a gradual increase

in λ_q during the torque scan: λ_q increases from ~ 2.8 mm at counter- I_p torque up to ~ 3.8 mm at co- I_p torque, as shown in figure 21(a). Strong correlation is observed between λ_q and edge turbulence intensity $\delta n/n$ at $\rho \sim 0.97$, as shown in figure 21(b).

Measured values of λ_q can be compared with L-mode scaling based on JET and ASDEX-Upgrade databases [60]:

$$\lambda_q (\text{mm}) = 1.58 \cdot B_{\text{tor}}^{-0.4} (\text{T}) P_{\text{SOL}}^{0.13} (\text{MW}) q_{95}^{0.73} R_0^{0.26} (m) \quad (10)$$

where B_{tor} is the toroidal magnetic field, $P_{\text{SOL}} = P_{\text{OH}} + P_{\text{NBI}} + P_{\text{ECH}} - P_{\text{rad,core}} - \frac{dW}{dt}$ is the power which goes to the scrape-off layer, q_{95} is the safety factor at $\Psi_N = 0.95$, and R_0 is the major radius of the plasma column. For the torque scan discharge #191379, we obtain a scaling value of $\lambda_q \approx 3.6$ mm, which is slightly below the maximum measured $\lambda_q \approx 3.8$ mm.

Another L-mode scaling can be found in [61], equation (7):

$$\lambda_q (\text{mm}) = 1.37 \cdot B_{\text{tor}}^{-0.55} (\text{T}) P_{\text{SOL}}^{0.2} (\text{MW}) q_{\text{cyl}}^{1.17} R_0^{0.1} (m) \quad (11)$$

where $q_{\text{cyl}} = \frac{\pi a^2 B_{\text{tor}} (1 + \kappa^2)}{\mu_0 I_p R_0}$ is the cylindrical safety factor. This scaling gives $\lambda_q \approx 4.18$ mm, which is larger than all measured values.

The HD model [8] gives the following estimation of heat flux decay length $\lambda_q \sim 2a\rho_p/R \approx 2.1$ mm, which is expectedly less than all the observed λ_q values, as this model does not take into account turbulent transport. For this estimation, we used the following parameters from DIII-D shot #191379: $a = 0.57$ m, $R = 1.71$ m, $T_{i,\text{sep}} = 60$ eV, $B_{\text{pol,sep}} = 0.364$ T.

E_r shear may inhibit the effect of edge turbulence on λ_q : during ECH/NBI power ramps, both $\delta n/n$ and E_r shear increase at the plasma edge, and we do not see any changes in λ_q . However, during the torque scan, the edge E_r shear is weak and, with the increase of $\delta n/n$, the mean radial velocity of density fluctuations $\langle v_r \rangle$ at $\rho = 0.97$ changes from negative to positive, indicating outward movement of ‘blobs’ across the separatrix. It should be noted that the shear of turbulence poloidal velocity v_{pol} weakens as well—at the strongest counter- I_p torque, the v_{pol} profile has some indication of shear at $\rho \sim 0.93$ (black curve in figure 20(b)), but with co- I_p torque the v_{pol} profiles are mostly flat at $\rho < 0.98$.

4. Summary

Turbulence properties of long-wavelength ion gyro-scale ($k\rho_i < 1$) density fluctuations are investigated at the edge of DIII-D L-mode plasmas in ECH/NBI power ramps and injected torque scan scenarios using BES. We find that broadband turbulent density fluctuations ($f \sim 7\text{--}130$ kHz) have a non-Gaussian distribution. The kurtosis of $\delta n/n$ is positive and increases with the increase of ECH/NBI power across the radial range of BES measurement $\rho = 0.88\text{--}1.1$. The skewness of $\delta n/n$ changes sign from negative at $\rho < 0.95\text{--}0.97$ to positive at outer radii. The velocimetry analysis of BES data shows inward radial velocity of density fluctuations in the area of negative skewness and outward radial velocity in the

area of positive skewness, which exhibits the inward movement of negative density fluctuations, ‘voids’, and the outward movement of positive fluctuations, ‘blobs’. Turbulence intensity flux $\langle \tilde{v}_r \tilde{n}^2 \rangle$ is directed inward inside the separatrix, which demonstrates inward turbulence spreading. The observations support a theory of ‘blob-void’ pair formation as the origin of turbulence spreading: inward spreading is associated with the inward movement of density ‘voids’.

Strong E_r shear may inhibit propagation of density ‘blobs’ across the separatrix and the effect of edge turbulence on the SOL heat flux decay length λ_q : during ECH/NBI power ramps, both $\delta n/n$ and E_r shear at the plasma edge increase with the injected power, only inward spreading is observed inside the last closed flux surface, and there are no changes in λ_q . During the torque scan, edge E_r shear is much weaker, and a strong correlation is observed between λ_q , injected torque T_{inj} , and turbulence amplitude $\delta n/n$ ($\rho = 0.97$). With the increase of $\delta n/n$, the mean radial velocity of density fluctuations $\langle v_r \rangle$ at $\rho = 0.97$ changes from negative to positive, which, in addition to positive skewness, indicates outward movement of ‘blobs’ across the separatrix.

Further work requires simulations that will help to gain a better understanding of the physics of turbulence spreading from the plasma edge and its effect on λ_q . Dedicated experiments are needed to measure the turbulence production ratio R_a in the SOL, and to study inward turbulence spreading from the plasma edge towards the core plasma, as well as the connection between $\delta n/n$, E_r shear, and λ_q broadening.

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BES and ECEI data analysis were performed using the OMFIT integrated modelling framework [42, 62, 63].

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Appendix. Derivation of model equation

To derive the equation for the evolution of the turbulence intensity in the SOL, we use the Hasegawa–Wakatani model. The model consists of two equations: the vorticity equation and the electron continuity equation [30]. The vorticity equation can be derived from the equation

$$\nabla \cdot \mathbf{J}_{pol} + \nabla \cdot \mathbf{J}_{PS} + \nabla_{\parallel} J_{\parallel} = 0, \quad (\text{A1})$$

where $\mathbf{J}_{pol}$, $\mathbf{J}_{PS}$, and $J_{\parallel}$ are polarization current, Pfirsch–Schlüter current, and parallel current, respectively.

The main contribution to the polarization current is from ions, so

$$\tilde{J}_{pol} = n_0 e \tilde{v}_{pol} = n_0 e \frac{m_i}{e} \frac{1}{B_0^2} \frac{d\tilde{E}_{\perp}}{dt} = -n_0 m_i \frac{1}{B_0^2} \frac{d\nabla_{\perp} \tilde{\varphi}}{dt}, \quad (\text{A2})$$

where $d_t = \partial_t + \tilde{\mathbf{v}}_{E \times B} \cdot \nabla$, and $\sim$ denotes the fluctuating part.

The Pfirsch–Schlüter current originates from the curvature drift: $\mathbf{v}_B = -\frac{2T_e}{eB_0} \frac{1}{R_c} (\hat{\mathbf{r}} \times \hat{\mathbf{b}})$, where R_c is the magnetic curvature radius. Thus, the Pfirsch–Schlüter current is

$$\tilde{J}_{PS} = -e \tilde{n} v_B = \tilde{n} \frac{2T_e}{B_0} \frac{1}{R_c} (\hat{\mathbf{r}} \times \hat{\mathbf{b}}). \quad (\text{A3})$$

The expression for $\tilde{J}_{\parallel}$ can be derived from the electron drift kinetic equation:

$$\tilde{J}_{\parallel} = \frac{v_{the}^2}{\nu_{ei}} n_0 e \nabla_{\parallel} \left(\frac{\tilde{n}}{n_0} - \frac{e\tilde{\varphi}}{T_e} \right). \quad (\text{A4})$$

Here, v_{the} is the electron thermal velocity and ν_{ei} is the electron–ion collision frequency.

Adopting the standard transformation [30]

$$\frac{x}{\rho_s} \rightarrow x, \quad t \omega_{ci} \rightarrow t, \quad \frac{v}{c_s} \rightarrow v, \quad \frac{e}{T_e} \varphi \rightarrow \varphi, \quad (\text{A5})$$

and dividing it by $n_0 e \omega_{ci}$, the vorticity equation (A1) can be rewritten into the following dimensionless form

$$\frac{d}{dt} \nabla_{\perp}^2 \tilde{\varphi} + \frac{2\rho_s}{R_c} \frac{1}{n_0} \frac{\partial \tilde{n}}{\partial y} = D_{\parallel} \nabla_{\parallel}^2 \left(\frac{\tilde{n}}{n_0} - \tilde{\varphi} \right). \quad (\text{A6})$$

Here, c_s is the ion sound speed, ω_{ci} is the ion gyrofrequency, $\rho_s = c_s/\omega_{ci}$ is the sound speed gyroradius, $D_{\parallel} = v_{the}^2/\nu_{ei} \cdot 1/(\omega_{ci}\rho_s^2)$ is the parallel diffusivity, and y is the poloidal direction (y lies on the flux surface). By using the same transformation (A5), the dimensionless continuity equation reads

$$\frac{1}{n_0} \frac{dn}{dt} + \frac{2\rho_s}{R_c} \frac{1}{n_0} \frac{\partial \tilde{n}}{\partial y} = D_{\parallel} \nabla_{\parallel}^2 \left(\frac{\tilde{n}}{n_0} - \tilde{\varphi} \right). \quad (\text{A7})$$

The Hasegawa–Wakatani equation system with curvature drive consists of equations (A6) and (A7). Note that in both equations n is still dimensional and thus has the dimension of plasma density.

In the SOL, the sheath boundary condition applies, so the electron response on the right-hand side of equations (A6) and (A7) should be replaced accordingly. The sheath parallel current is

$$(J_{\parallel})_{sh} = n_0 e c_s \left(1 - e^{-e(\varphi - \varphi_f)/T_e} \right), \quad (\text{A8})$$

where φ_f is the floating potential. Then with $\frac{e}{T_e} \varphi \rightarrow \varphi$ [31]:

$$\nabla_{\parallel} \tilde{J}_{\parallel} \approx \frac{2n_0 e^2 c_s \tilde{\varphi}}{L_{\parallel} T_e} \rightarrow \frac{2n_0 e c_s}{L_{\parallel}} \tilde{\varphi}, \quad (\text{A9})$$

where $L_{\parallel}$ is the parallel connection length. During the derivation of the dimensionless vorticity equation (A6) we divided the equation by $n_0 e \omega_{ci}$, so to get the dimensionless $\nabla_{\parallel} \tilde{J}_{\parallel}$ we should do the same with equation (A9): $\nabla_{\parallel} \tilde{J}_{\parallel} = 2\rho_s \tilde{\varphi}/L_{\parallel}$.

For the continuity equation in the SOL, the parallel dynamics is defined by the parallel loss $\nabla_{\parallel} \tilde{J}_{\parallel} = \nabla_{\parallel} \tilde{n} v_{\parallel} = 2\tilde{n} e c_s/L_{\parallel}$ [31]. In the dimensionless form it becomes $\nabla_{\parallel} \tilde{J}_{\parallel} = 2\tilde{n} \rho_s/n_0 L_{\parallel}$.

The Hasegawa–Wakatani equations with the sheath boundary condition are

$$\frac{d}{dt} \nabla_{\perp}^2 \tilde{\varphi} + \frac{2\rho_s}{R_c} \frac{1}{n_0} \frac{\partial \tilde{n}}{\partial y} = \frac{2\rho_s}{L_{\parallel}} \tilde{\varphi} \quad (\text{A10})$$

$$\frac{1}{n_0} \frac{dn}{dt} + \frac{2\rho_s}{R_c} \frac{1}{n_0} \frac{\partial \tilde{n}}{\partial y} = -\frac{2\rho_s}{L_{\parallel}} \frac{\tilde{n}}{n_0}. \quad (\text{A11})$$

To obtain the evolution of the turbulence intensity in the SOL, we rewrite (A11) in detail:

$$\frac{1}{n_0} \left(\frac{\partial}{\partial t} + \tilde{\mathbf{v}} \cdot \nabla \right) (n_0 + \tilde{n}) + \frac{2\rho_s}{R_c} \frac{1}{n_0} \frac{\partial \tilde{n}}{\partial y} = -\frac{2\rho_s}{L_{\parallel}} \frac{\tilde{n}}{n_0} \Rightarrow \quad (\text{A12})$$

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\tilde{n}}{n_0} + \tilde{v}_r \frac{\partial}{\partial r} \frac{\tilde{n}}{n_0} + \tilde{v}_y \frac{\partial}{\partial y} \frac{\tilde{n}}{n_0} + \tilde{v}_r \frac{\partial}{\partial r} \ln n_0 + \frac{2\rho_s}{R_c} \frac{\partial}{\partial y} \frac{\tilde{n}}{n_0} \\ = -\frac{2\rho_s}{L_{\parallel}} \frac{\tilde{n}}{n_0}. \end{aligned} \quad (\text{A13})$$

Then in (A13), making $\tilde{n}$ dimensionless with the transformation $\tilde{n}/n_0 \rightarrow \tilde{n}$, multiplying by $\tilde{n}$, and taking an average over time and y ($\partial/\partial y$ terms vanish) yields the following:

$$\frac{1}{2} \frac{\partial \langle \tilde{n}^2 \rangle}{\partial t} + \frac{1}{2} \frac{\partial \langle \tilde{v}_r \tilde{n}^2 \rangle}{\partial r} + \langle \tilde{n} \tilde{v}_r \rangle \frac{\partial}{\partial r} \ln n_0 = -\frac{2\rho_s}{L_{\parallel}} \langle \tilde{n}^2 \rangle. \quad (\text{A14})$$

Equation (A14) is dimensionless, and moving back to dimensional form and keeping the time derivative on the left-hand side gives equation (1):

$$\frac{1}{2} \frac{\partial \langle \tilde{n}^2 \rangle}{\partial t} = -\frac{1}{2} \frac{\partial \langle \tilde{v}_r \tilde{n}^2 \rangle}{\partial r} - \langle \tilde{n} \tilde{v}_r \rangle \frac{\partial n_0}{\partial r} - \frac{2c_s}{L_{\parallel}} \langle \tilde{n}^2 \rangle. \quad (\text{A15})$$

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